(b) Explain principle of induction heating in detail. 7

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(c) A slab of insulating material 0·015 m^2 in area and 0·01 m in thickness has relative permittivity of 5 and power factor of 0·05 and is to be heated by dielectric heating. The power required is 400 W at 30 MHz. Determine the voltage required and the resulting current that will flow through the material. If the voltage were to be limited to 650 V. What would be the value of the frequency for the same power requirement. 7

(d) Classify Resistance weilding and explain in brief. 7

Unit-V

5. (a) Explain origin of noise. 2

(b) Briefly explain with the help of block diagram operation of online and offline UPS. 7

(c) What are the different types of noises? Explain in brief. 7

(d) Explain the principle of operation of constant voltage transformer. 7

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B. E. (Eighth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Et & T Branch)

INDUSTRIAL & POWER ELECTRONICS

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Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.*

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Unit-I

1. (a) Define input line regulation factor and load regulation factor of regulated power supply. 2

(b) Explain with a neat circuit diagram the operation of buck-boost regulator. 7

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(c) Explain with the help of neat circuit diagram the operation of transistorized series voltage regulator. 7

(d) Explain three terminal voltage regulator IC LM340 and prove that : 7

$$V_{out} = V_{reg}\left(1+\frac{R_2}{R_1}\right)$$

$$I_L = \frac{V_{reg}}{R_1}$$

Unit-II

2. (a) Define holding current and latching current. 2

(b) For SCR gate cathode characteristics is given by $V_g = 1+10Ig$. Gate source voltage is a rectangular pulse of 15 V with 20 μsec duration for an average power dissipation of 0·3 watt and peak gate drive power of 5 watt. Calculate : 7

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(i) Resistance to be connected in series with SCR gate.

(ii) Triggering frequency

(iii) Duty cycle of the triggering pulse.

(c) Explain class C commutation method of SCR. 7

(d) Explain dynamic characteristic of SCR in detail. 7

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Unit-III

3. (a) What do you mean by chopper and where it is used? 2

(b) Explain single phase mid point cyclo-converter with the help of circuit diagram. 7

(c) An voltage commuted step down chopper operates at 500 Hz from 48 V battery and feeds a level load of 5 Amp. Calculate values of commutating components. The turn off time for main SCR is 80 μsec while that for auxillary SCR is 40 μs. 7

(d) Describe Jones Chopper in brief. 7

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Unit-IV

4. (a) What is Skin effect? 2